

Experimental Instructions for “Belief Disagreement, Instability, and Uncertainty under Stochastic Recall”

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1 Field Study 1: Stock Market Returns

1.1 Part 1

Section 1: Instruction Overview

Thank you for taking part in this online study.

Part 1 (today) and Part 2 (tomorrow). This study consists of **two parts**. Part 1 takes place now, and Part 2 will take place tomorrow at around the same time. Part 1 and Part 2 are paid separately, and you will receive a participation payment for each part. By completing both parts of the study, you will also have a chance to earn an **additional bonus of \$30**.

Instruction. In the next section, you will see a series of **stock market returns** from 2025, presented one at a time. Please review each screen carefully. You do not need to take notes. Simply view the information as it is presented. Some of your responses will be selected for a bonus payment. Please read the information carefully, as accurate answers may increase your bonus.

Section 2: Stock market returns display

The following twelve monthly returns are presented sequentially, one per screen:

- January 2025 stock market return: 2.7%
- February 2025 stock market return: -1.42%
- March 2025 stock market return: -5.75%
- April 2025 stock market return: -0.76%
- May 2025 stock market return: 6.15%
- June 2025 stock market return: 4.96%
- July 2025 stock market return: 2.17%
- August 2025 stock market return: 1.91%
- September 2025 stock market return: 3.53%
- October 2025 stock market return: 2.27%
- November 2025 stock market return: 0.13%
- December 2025 stock market return: -0.05%

Q1. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? Your answer will not affect your bonus payment for this study. Please answer truthfully.

- Yes
- No

Section 3: Estimate and recalls

Q2. Attention check. This is a question to check whether you are paying attention and reading the questions carefully. Please select both “strongly disagree” and “strongly agree” to continue. Otherwise, you will leave the survey.

- Strongly agree
- Agree
- Neither strongly agree or disagree
- Disagree
- Strongly disagree

Q3. Your estimate

Based on the information you just received, what is your best estimate of the **average monthly stock market return in 2025?**

[Bonus: After the study, answers will be compared to the true average of the data series you just observed. The 10% of participants with the most accurate answers will each receive a \$1 bonus.]

Average monthly stock market return: _____%

Q4. Recalls

Please enter any stock market return you remember seeing earlier. Please recall as many as you can.

- January 2025: _____%
- February 2025: _____%
- March 2025: _____%
- April 2025: _____%
- May 2025: _____%
- June 2025: _____%
- July 2025: _____%
- August 2025: _____%
- September 2025: _____%
- October 2025: _____%
- November 2025: _____%
- December 2025: _____%

Section 4: Background information

Q5. What is the state of your primary residence? *[Dropdown list]*

Q6. How would you describe your present employment situation?

- Working full-time (for someone or self-employed)
- Working part-time (for someone or self-employed)
- Not working, but would like to work
- Not actively looking for work
- Student, at school or in training
- Other

Q7. Which category represents your total household pre-tax income during the past 12 months?

- Less than \$10,000
- \$10,000 to \$19,999
- \$20,000 to \$29,999
- \$30,000 to \$39,999
- \$40,000 to \$49,999
- \$50,000 to \$59,999
- \$60,000 to \$74,999
- \$75,000 to \$99,999
- \$100,000 to \$149,999
- \$150,000 to \$199,999
- \$200,000 or more

Q8. Does your household hold any stock market investments?

- Yes
- No

Q9. What is your gender?

- Male
- Female
- Other
- Prefer not to answer

Q10. What is your age? *[Dropdown list]*

Q11. What is the highest level of school you have completed, or the highest degree you have received?

- Less than high school

- High school diploma
- Some college but no degree
- Bachelor's Degree or higher

1.2 Part 2 (one day after Part 1)

Welcome

Thank you for participating in this study. This study consists of two parts. Part 1 took place yesterday and Part 2 takes place now. To complete the study, you will need to read all instructions carefully.

Part 2 – Instruction. Yesterday in Part 1, you received a series of stock market returns in 2025. This means **all of the information you received in Part 1 yesterday is relevant** for making your decisions on the following screens. You have a chance to win an additional bonus of up to \$30 by completing this part of survey.

Q1. Your estimate

Based on the information you received yesterday in Part 1, what is your best estimate of the average monthly stock market return in 2025?

[Bonus: After the study, answers will be compared to the true average of the data series you just observed. The 10% of participants with the most accurate answers will each receive a \$1 bonus.]

Average monthly stock market return: _____%

Q2. Recalls

Please enter any stock market return you remember seeing yesterday in Part 1. Please recall as many as you can.

- January 2025: _____%
- February 2025: _____%
- March 2025: _____%
- April 2025: _____%
- May 2025: _____%
- June 2025: _____%
- July 2025: _____%
- August 2025: _____%
- September 2025: _____%
- October 2025: _____%
- November 2025: _____%
- December 2025: _____%

2 Field Study 2: Inflation Rates

2.1 Part 1

Section 1: Instruction Overview

Thank you for taking part in this online study.

Part 1 (today) and Part 2 (tomorrow). This study consists of two parts. Part 1 takes place now, and Part 2 will take place tomorrow at around the same time. Part 1 and Part 2 are paid separately, and you will receive a participation payment for each part. By completing both parts of the study, you will also have a chance to earn an additional bonus of \$30.

Instruction. In the next section, you will see a series of **inflation rates** over the past 10 years presented one at a time. Please review each screen carefully. You do not need to take notes. Simply view the information as it is presented. Some of your responses will be selected for a bonus payment. Please read the information carefully, as accurate answers may increase your bonus.

Section 2: Inflation rates display

The following ten annual inflation rates are presented sequentially, one per screen:

- Inflation rate in 2016: 2.05%
- Inflation rate in 2017: 2.13%
- Inflation rate in 2018: 2%
- Inflation rate in 2019: 2.32%
- Inflation rate in 2020: 1.32%
- Inflation rate in 2021: 7.16%
- Inflation rate in 2022: 6.41%
- Inflation rate in 2023: 3.32%
- Inflation rate in 2024: 2.87%
- Inflation rate in 2025: 2.65%

Q1. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? Your answer will not affect your bonus payment for this study. Please answer truthfully.

- Yes
- No

Section 3: Estimate and recalls

Q2. Attention check. This is a question to check whether you are paying attention and reading the questions carefully. Please select both “strongly disagree” and “strongly agree” to continue. Otherwise, you will leave the survey.

- Strongly agree
- Agree
- Neither strongly agree or disagree
- Disagree
- Strongly disagree

Q3. Your estimate

Based on the information you just received, what is your best estimate of the average annual inflation rate over the past 10 years?

[Bonus: After the study, answers will be compared to the true average of the data series you just observed. The 10% of participants with the most accurate answers will each receive a \$1 bonus.]

Average inflation rate: _____%

Q4. Recalls

Please enter any inflation rate you remember seeing earlier. Please recall as many as you can.

- Inflation rate in 2016: _____%
- Inflation rate in 2017: _____%
- Inflation rate in 2018: _____%
- Inflation rate in 2019: _____%
- Inflation rate in 2020: _____%
- Inflation rate in 2021: _____%
- Inflation rate in 2022: _____%
- Inflation rate in 2023: _____%
- Inflation rate in 2024: _____%
- Inflation rate in 2025: _____%

Section 4: Background information

The Section 4 background-information questions (state, employment, income, stock investments, gender, age, education) are identical to those in Field Study 1, Q5–Q11.

2.2 Part 2 (one day after Part 1)

Welcome

Thank you for participating in this study. This study consists of two parts. Part 1 took place yesterday and Part 2 takes place now. To complete the study, you will need to read all instructions carefully.

Part 2 – Instruction. Yesterday in Part 1, you received a series of inflation rates over the past 10 years. This means all of the information you received in Part 1 yesterday is relevant for making your decisions on the following screens. You have a chance to win an additional bonus of up to \$30 by completing this part of survey.

Q1. Your estimate

Based on the information you received yesterday in Part 1, what is your best estimate of the average annual inflation rate over the past 10 years?

[Bonus: After the study, answers will be compared to the true average of the data series you just observed. The 10% of participants with the most accurate answers will each receive a \$1 bonus.]

Average inflation rate: _____%

Q2. Recalls

Please enter any inflation rate you remember from yesterday in Part 1. Please recall as many as you can.

- Inflation rate in 2016: _____%
- Inflation rate in 2017: _____%
- Inflation rate in 2018: _____%
- Inflation rate in 2019: _____%
- Inflation rate in 2020: _____%
- Inflation rate in 2021: _____%
- Inflation rate in 2022: _____%
- Inflation rate in 2023: _____%
- Inflation rate in 2024: _____%
- Inflation rate in 2025: _____%

3 Field Study 3: Tariff Policy Headlines

3.1 Part 1

Section 1: Instruction Overview

Thank you for taking part in this online study.

Part 1 (today) and Part 2 (tomorrow). This study consists of two parts. Part 1 takes place now, and Part 2 will take place tomorrow at around the same time. Part 1 and Part 2 are paid separately, and you will receive a participation payment for each part. By completing both parts of the study, you will also have a chance to earn an additional bonus of \$30.

Instruction. In the next section, you will see a series of **headlines from opinion articles about Donald Trump’s tariff policy** taken from some latest newspapers, presented one at a time. Please review each screen carefully. You do not need to take notes. Simply view the information as it is presented. Some of your responses will be selected for a bonus payment. Please read the information carefully, as accurate answers may increase your bonus.

Section 2: Headlines display

The following eleven headlines are presented sequentially, one per screen. The article tone (positive/negative) is also displayed.

- “Trump’s Tariffs Risk Economic Turbulence and Voter Backlash” (*negative*)
- “Trump’s Tariff Revenue Could Help Pay Down the Nation’s \$37.2T Debt” (*positive*)
- “Trump’s Self-inflicted Tariff Crisis Sparks Confusion, Chaos and Questions of Competence” (*negative*)
- “Trump’s Tariffs Would Smother His Economic Successes” (*negative*)
- “Economist Says Trump’s Tariffs Are a ‘Win-win Strategy’ to Negotiate Trade” (*positive*)
- “Tariffs Can Help US Workers. But Trump’s Doing Them All Wrong” (*negative*)
- “Trump’s New Tariff War Is Perfectly Rational” (*positive*)
- “Trump’s Tariff Victory Is Not What It Seems” (*negative*)
- “Trump’s Tariff Turbulence Is Worse Than Anyone Imagined” (*negative*)
- “Trump Tariffs Will Lead to World Prosperity, Economist Says” (*positive*)
- “Look on the Bright Side of Trump’s Global Tariffs” (*positive*)

Q1. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? Your answer will not affect your bonus payment for this study. Please answer truthfully.

- Yes
- No

Section 3: Opinion and recalls

Q2. Attention check. This is a question to check whether you are paying attention and reading the questions carefully. Please select both “strongly disagree” and “strongly agree” to continue. Otherwise, you will leave the survey.

- Strongly agree
- Agree
- Neither strongly agree or disagree
- Disagree
- Strongly disagree

Q3. Your opinion

[Randomly flip the sequence of options] Based on the opinion article headlines you just read, how would you describe the overall sentiment of these articles toward Trump’s tariffs?

- Very negative
- Somewhat negative
- Somewhat positive
- Very positive

Q4. Recalls

[Randomly display the order of positive and negative] Based on the opinion article headlines you just read, how many times was a positive article shown to you, and how many times was a negative article shown to you?

Bonus Payment: If your answers are correct, you will receive a £1 bonus.

- Positive news: _____
- Negative news: _____

Section 4: Background information

The state, employment, income, stock investments, gender, age, and education questions (Q5–Q11) are identical to those in Field Study 1.

Q12. Do you consider yourself a(n):

- Democrat
- Republican
- Independent

- None of the above

Q12a. *(Conditional on choosing Democrat or Republican)* Are you a strong or moderate Democrat/Republican?

- Strong
- Moderate

Q12b. *(Conditional on choosing Independent or None of the above)* Do you consider yourself closer to the:

- Democratic Party
- Republican Party

3.2 Part 2 (one day after Part 1)

Welcome

Thank you for participating in this study. This study consists of two parts. Part 1 took place yesterday and Part 2 takes place now. To complete the study, you will need to read all instructions carefully.

Part 2 – Instruction. Yesterday in Part 1, you received a series of opinion article headlines about Donald Trump’s tariffs. This means all of the information you received in Part 1 yesterday is relevant for making your decisions on the following screens. You have a chance to win an additional bonus of up to \$30 by completing this part of survey.

Q1. Your opinion

[Randomly flip the sequence of options] Based on the opinion article headlines you read yesterday in Part 1, how would you describe the overall sentiment of these articles toward Trump’s tariffs?

- Very negative
- Somewhat negative
- Somewhat positive
- Very positive

Q2. Recalls

[Randomly display the order of positive and negative] Based on the opinion article headlines you read yesterday in Part 1, how many times was a positive article shown to you, and how many times was a negative article shown to you?

Bonus Payment: If your answers are correct, you will receive a £1 bonus.

- Positive news: _____
- Negative news: _____

4 Field Study 4: Electric Vehicle Company Headlines

4.1 Part 1

Section 1: Instruction Overview

Thank you for taking part in this online study.

Part 1 (today) and Part 2 (tomorrow). This study consists of two parts. Part 1 takes place now, and Part 2 will take place tomorrow at around the same time. Part 1 and Part 2 are paid separately, and you will receive a participation payment for each part. By completing both parts of the study, you will also have a chance to earn an additional bonus of \$30.

Instruction. In the next section, you will see a series of recent **news article headlines about a major electric vehicle company (Tesla)**, presented one at a time. Please review each screen carefully. You do not need to take notes. Simply view the information as it is presented. Some of your responses will be selected for a bonus payment. Please read the information carefully, as accurate answers may increase your bonus.

Section 2: Headlines display

The following eleven headlines are presented sequentially, one per screen. The headline tone (positive/negative) is also displayed.

- “Tesla Remains a Leader in Electric Vehicle Innovation” (*positive*)
- “Tesla’s Car Sales Rose 7% as U.S. E.V. Tax Credit Was Ending” (*positive*)
- “Tesla Profit Falls, Hurt by Plunging EV Sales” (*negative*)
- “Tesla Benefits From Long-run Growth in EV Adoption” (*positive*)
- “Tesla’s Brand Loyalty Continues to Attract Customers” (*positive*)
- “Tesla Reports Falling Profit as Car Sales Slow” (*negative*)
- “Tesla’s Energy Business Supports Diversification Beyond Cars” (*positive*)
- “Tesla Reports 20% Drop in Automotive Revenue” (*negative*)
- “Tesla Shares Slide as Musk Feuds With Trump” (*negative*)
- “Tesla’s Manufacturing Focus Could Improve Efficiency” (*positive*)
- “China’s BYD Surpasses Tesla as World Leader in Electric Car Sales” (*negative*)

Q1. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? Your answer will not affect your bonus payment for this study. Please answer truthfully.

- Yes
- No

Section 3: Opinion and recalls

Q2. Attention check. This is a question to check whether you are paying attention and reading the questions carefully. Please select both “strongly disagree” and “strongly agree” to continue. Otherwise, you will leave the survey.

- Strongly agree
- Agree
- Neither strongly agree or disagree
- Disagree
- Strongly disagree

Q3. Your opinion

[Randomly flip the sequence of options] Based on the news headlines you just read, how would you describe the overall sentiment of the news coverage toward Tesla?

- Very negative
- Somewhat negative
- Somewhat positive
- Very positive

Q4. Recalls

[Randomly display the order of positive and negative] Based on the news headlines you just read, how many times was positive news shown to you, and how many times was negative news shown to you?

Bonus Payment: If your answers are correct, you will receive a £1 bonus.

- Positive news: _____
- Negative news: _____

Section 4: Background information

The state, employment, income, stock investments, gender, age, and education questions (Q5–Q11) are identical to those in Field Study 1.

4.2 Part 2 (one day after Part 1)

Welcome

Thank you for participating in this study. This study consists of two parts. Part 1 took place yesterday and Part 2 takes place now. To complete the study, you will need to read all instructions carefully.

Part 2 – Instruction. Yesterday in Part 1, you received a series of news headlines about a major electric vehicle company (Tesla). This means all of the information you received in

Part 1 yesterday is relevant for making your decisions on the following screens. You have a chance to win an additional bonus of up to \$30 by completing this part of survey.

Q1. Your opinion

[Randomly flip the sequence of options] Based on the news headlines you read yesterday in Part 1, how would you describe the overall sentiment of the news coverage toward Tesla?

- Very negative
- Somewhat negative
- Somewhat positive
- Very positive

Q2. Recalls

[Randomly display the order of positive and negative] Based on the news headlines you read yesterday in Part 1, how many times was positive news shown to you, and how many times was negative news shown to you?

Bonus Payment: If your answers are correct, you will receive a £1 bonus.

- Positive news: _____
- Negative news: _____

5 Belief Disagreement Experiment (Experiment 1)

Prolific ID

What is your Prolific ID? Please note that this response should auto-fill with the correct ID.

5.1 Section 1: Pre-experiment instruction

Overview. Thank you for taking part in this online experiment. Please complete the study in one sitting, without taking breaks in between, and keep your attention on the survey until the end.

Comprehension checks. On next pages, you will receive detailed instruction and information about the experiment. Afterwards, you will be asked to complete a few comprehension check questions. You can only participate in the experiment if you correctly complete all comprehension check questions!

Instruction. In this experiment, you will see a hypothetical company with no connection to real-world firms. The company can be either good or bad. A good company means its business goes well, while a bad company means its business goes poorly. The computer decides whether it is good or bad by flipping a fair coin.

- If Heads comes up, the company is good (probability 50%)
- If Tails comes up, the company is bad (probability 50%)

Important: You will not be told the outcome of the coin flip. This means you will not know directly whether the company is good or bad. Instead, you will receive multiple pieces of news about the company. These news items are designed to help you judge whether the company is more likely to be good or bad.

The news. To illustrate the process, imagine that a box contains 100 pieces of news for the company. The computer randomly draws items from this box to show you.

- A **good** company has a box with 65 positive news items and 35 negative news items.
- A **bad** company has a box with 35 positive news items and 65 negative news items.

This means that for a good company, the probability of drawing positive news is 65%, while for a bad company it is only 35%.

After each draw, the news item is placed back into the box before the next draw. This means that all news are drawn with replacement, so each draw is **independent and equally informative**.

Summary: In this experiment, you will never know with certainty whether the company is good or bad. However, you will receive helpful information. While the news are prone to some error, they are on average helpful in estimating whether a company is good or bad. For example, if you receive multiple positive news for a company, this is of course a stronger indication that the company is actually good than if you only receive one positive news.

Communication of the news. For the company, you will receive multiple pieces of news sequentially on separate screens. Each piece of news (positive or negative) will be communicated with an **event**. An event consists of a short story that explains why positive or negative news occurred. An illustrative image may appear alongside the story, but the image itself does NOT contain additional information beyond the text.

Events are specific to the type of news. The same event cannot appear for both positive and negative news. Example: You might receive a negative piece of news communicated with the event “Factory burns down.” You would then see a short description of the fire (and possibly an image). This particular event could only appear for negative news.

Important: Any given event and the corresponding news could appear MULTIPLE TIMES for the company. In such cases, you would have to take into account these news multiple times (and equally strongly), regardless of whether you have seen that event before. For example, it could happen that the company’s factory burns down twice. You would then have to take into account these news twice. Events are also type-specific. An event used for positive news will never be used for negative news, and vice versa. Ultimately, all that matters for you is how many times you receive positive and negative news, not which events cause these news or whether the same event appears multiple times.

Your decisions. After you have received these pieces of news about the company, you will be asked to decide whether you think the company is good or bad.

Your payment. At the end of the study, if your guess for the company is correct, you will earn a \$0.5 bonus. If your guess is incorrect, you will not earn the bonus. In addition, you will receive a fixed participation payment for completing the study.

In the next page, there will be some comprehension checks. You can only continue the experiment if you correctly complete all comprehension check questions. If you are still unsure about the instructions, please go back and read them carefully before answering the comprehension check questions.

5.2 Section 2: Comprehension checks

[Option sequence is randomized.] There are seven comprehension check questions that will test your understanding of the instructions. You can only participate in the experiment if you correctly answer all questions below.

Q1. Is the following statement correct? “The more positive news I receive for a company, the more likely it is that the company is good.”

- Yes
- No

Q2. Suppose one event is given by “Factory burns down.” Which of the following statements is correct?

- The event can never occur multiple times.
- The event can occur multiple times as positive and negative news.
- The event can occur multiple times but only as negative news.

Q3. Which of the following statements is correct?

- If I have already seen an event that is associated with positive news, then I know I have already received a positive piece of news for this company.
- The same event can be associated with both positive and negative news.

Q4. Suppose that you have received two negative news items for a company, both communicated with the same event. Which of the following statements is correct?

- Only one of the two news should be taken into account because they were communicated with the same event.
- I need to take both pieces of news into account because both were randomly and independently determined.

Q5. Suppose that you have received two positive news items for a company. Which of the following statements is correct?

- I should take them into account more strongly if they were communicated with the same event than if they were communicated with different events.
- I should take them into account less strongly if they were communicated with the same event than if they were communicated with different events.
- It doesn't matter; I should treat both equally, regardless of whether they come from the same event or different events.

Q6. Suppose that a company is good and that the first news item about this company was positive. What does this imply for the probability that the next piece of news will also be positive?

- The probability is the same as before, since the computer draws independently with replacement.
- The probability is higher than before.
- The probability is lower than before.

Q7. Do images shown alongside the text (story) provide additional information about whether the company is good or bad?

- Yes, the image itself contains important information.
- No, the image is only illustrative; all information comes from the text (story).

5.3 Section 3: Group randomization and news delivery

Instruction. First, on the following pages, you will now sequentially receive multiple news about the company. These news items will be shown to you on separate screens. You can only review each news once. After moving forward, you will NOT be able to go back to earlier news items. Please pay close attention to each news item before proceeding to the next one.

Reminder: At the end of the study, if your guess for that company is correct, you will earn a \$0.50 bonus payment; if it is incorrect, you will not earn the bonus.

[Respondents are randomized into one of the following 9 groups.]

Group structure

- **Group 1:** 18 positive news and 2 negative news.
- **Group 2:** 14 positive news and 6 negative news.
- **Group 3:** 11 positive news and 9 negative news.

- **Group 4:** 6 positive news and 14 negative news.
- **Group 5:** 2 positive news and 18 negative news.
- **Group 6 (Stop-and-Think):** 11 positive news and 9 negative news; complete a structured recall task *before* belief elicitation.
- **Group 7 (Monetary Incentives):** 11 positive news and 9 negative news; complete a structured recall task *before* belief elicitation; additional surprise compensation tied to the accuracy of their recall responses.
- **Group 8 (Positive Cue):** 14 positive news and 6 negative news; a visual cue associated with *positive* news enhances recall.
- **Group 9 (Negative Cue):** 14 positive news and 6 negative news; a visual cue associated with *negative* news enhances recall.

News sequences across treatment groups

The table below presents the sequence of 20 news signals assigned to each treatment group. Each signal is positive (P) or negative (N). Groups G6 and G7 receive the same news sequence as G3; Groups G8 and G9 receive the same news sequence as G2.

Position	G1	G2	G3	G4	G5
1	P	P	N	P	N
2	P	P	P	N	N
3	P	N	N	N	N
4	P	P	P	P	N
5	N	P	P	N	N
6	P	P	N	N	N
7	P	P	P	N	P
8	P	N	N	N	N
9	P	P	P	P	N
10	P	P	P	N	N
11	P	P	N	N	N
12	N	N	P	N	N
13	P	P	N	P	N
14	P	P	P	N	P
15	P	N	N	P	N
16	P	P	P	N	N
17	P	N	N	N	N
18	P	P	P	P	N
19	P	P	N	N	N
20	P	N	P	N	N

Example of a positive news item:



The company successfully launched a new product line that quickly attracted strong customer demand. The rollout went smoothly, and sales exceeded expectations in the first months.

The news for the company is positive.

Example of a negative news item:



A major customer cancelled its contract with the company, citing repeated delivery delays and quality concerns. The cancellation will significantly reduce the company's sales and production in the coming months.

The news for the company is negative.

Q8. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? Your answer will not affect your bonus payment for this study. Please answer truthfully.

- Yes
- No

5.4 Section 4: Belief elicitation and recall

Branch A: If assigned to Groups 1 to 5

Q9. Your decision. *[Option sequence is randomized.]* Based on the news items you just received about the company, what is your best guess about the type of this company? If your guess is correct, you will earn a \$0.50 bonus.

- Good
- Bad

[Randomly shown one of the questions below.]

Q10v1. What do you think is the **probability** that the company is “good”? Slider 0–100, labeled “Probability of being good.”

Q10v2. What do you think is the **probability** that the company is “bad”? Slider 0–100, labeled “Probability of being bad.”

Q11. Recalls. *[Sequence is randomized.]* Based on the news items you just received about the company, how many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____

- Negative news: _____

Q12. Item-position recall. Thinking back to the news items you just received, can you remember what **type of news** the 1st, 5th, 15th, and last items were?

For each of the four positions (1st, 5th, 15th, last) choose one of: Positive news / Negative news / I don't remember.

Branch B: If assigned to Groups 6 and 7

Q9 (Group 6). Recalls. *[Sequence is randomized.]* Based on the news items you just received about the company, how many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Q9 (Group 7). Incentivized recall. *[Option sequence is randomized.]* *Good news!* You have been selected to enter a special recall challenge. If you can accurately recall how many times positive and negative news items were shown to you for the company just now, you will earn an extra bonus of \$0.50!

Based on the news items you just received about the company, how many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Q10. Your decision. *[Option sequence is randomized.]* Based on the news items you just received about the company, what is your best guess about the type of this company? If your guess is correct, you will earn a \$0.50 bonus.

- Good
- Bad

[Randomly shown one of the questions below.]

Q11v1. What do you think is the probability that the company is “good”? Slider 0–100.

Q11v2. What do you think is the probability that the company is “bad”? Slider 0–100.

Q12. Item-position recall. Thinking back to the news items you just received, can you remember what type of news the 1st, 5th, 15th, and last items were? (Positive news / Negative news / I don't remember.)

Branch C: If assigned to Groups 8 and 9

Q9 (Group 8). *[Showing image associated with the positive news; option sequence is randomized.]* Based on the news items you just received about the company, what is your best guess about the type of this company? If your guess is correct, you will earn a \$0.50 bonus.

- Good
- Bad

Q9 (Group 9). *[Showing image associated with the negative news; option sequence is randomized.]* Based on the news items you just received about the company, what is your

best guess about the type of this company? If your guess is correct, you will earn a \$0.50 bonus.

- Good
- Bad

[Randomly shown one of the questions below.]

Q10v1. What do you think is the probability that the company is “good”? Slider 0–100.

Q10v2. What do you think is the probability that the company is “bad”? Slider 0–100.

Q11. Recalls. *[Sequence is randomized.]* Based on the news items you just received about the company, how many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Q12. Item-position recall. Thinking back to the news items you just received, can you remember what type of news the 1st, 5th, 15th, and last items were? (Positive news / Negative news / I don’t remember.)

5.5 Section 5: Patience and risk-taking characteristics

Q13. Today vs. future. How willing or unwilling are you to give up something that is beneficial for you today in order to benefit more from that in the future? Please use a scale from 0 to 10, where 0 means “completely unwilling to give up a benefit today for future” and 10 means you are “completely willing to give up a benefit today for future”.

Q14. Staircase time preference. Suppose you were given the choice between receiving a payment today or a payment in 12 months. We will now present to you five situations. The payment today is the same in each of these situations. The payment in 12 months is different in every situation. For each of these situations we would like to know which you would choose.

Note: Please assume there is no inflation, i.e., future prices are the same as today’s prices.

Please consider the following: Would you rather receive 100 dollars today or x dollars in 12 months? The precise sequence of questions is given by the staircase tree logic in Figure 1.

Q15. Willingness to take risks. Please tell me, in general, how willing or unwilling you are to take risks. Please use a scale from 0 to 10, where 0 means “completely unwilling to take risks” and 10 means you are “completely willing to take risks”.

Q16. Staircase risk task. Please imagine the following situation. You can choose between a sure payment of a particular amount of money, or a draw, where you would have an equal chance of getting \$300 or getting nothing. We will present to you five different situations.

What would you prefer: a draw with a 50 percent chance of receiving amount x, and the same 50 percent chance of receiving nothing, or the amount of y as a sure payment? The precise sequence of questions is given by the staircase tree logic in Figure 2.

5.6 Section 6: Financial knowledge and demographics

Q17. Suppose you put \$100 into a saving account and the interest rate is fixed at 3% per year for the next 5 years. After 5 years, how much do you think you would have in the account if you left the money to grow?

- more than \$115
- exactly \$115
- less than \$115
- don't know

Q18. Imagine that the interest rate on your saving account is 3% per year and inflation is 2% per year. After 1 year, how much would you be able to buy with the money in this account, compared to today?

- more than today
- exactly the same as today
- less than today
- don't know

Q19. Please indicate whether the statement is true or false: “buying a stock mutual fund usually provides safer return than buying a single company's stock”.

- true
- false
- don't know

Background information

Q20. What is the state of your primary residence? *[Dropdown list]*

Q21. How would you describe your present employment situation?

- Working full-time (for someone or self-employed)
- Working part-time (for someone or self-employed)
- Not working, but would like to work
- Not actively looking for work
- Student, at school or in training
- Other (please specify) _____

Q22. Which of the following best characterizes your household?

- Own our house/apartment without a mortgage
- Own our house/apartment and have a fixed-rate mortgage
- Own our house/apartment and have a variable-rate mortgage
- Rent our house/apartment

- Other

Q23. Which category represents your total household pre-tax income during the past 12 months?

- Less than \$10,000
- \$10,000 to \$19,999
- \$20,000 to \$29,999
- \$30,000 to \$39,999
- \$40,000 to \$49,999
- \$50,000 to \$59,999
- \$60,000 to \$74,999
- \$75,000 to \$99,999
- \$100,000 to \$149,999
- \$150,000 to \$199,999
- \$200,000 or more

Q24. Does your household hold any stock market investments?

- Yes
- No

Q25. What is your gender?

- Male
- Female
- Other
- Prefer not to answer

Q26. What is your age? *[Dropdown list]*

Q27. What is the highest level of school you have completed, or the highest degree you have received?

- Less than high school
- High school diploma
- Some college but no degree
- Bachelor's Degree or higher

6 Belief Instability Study (Experiments 2–3)

Prolific ID

What is your Prolific ID? Please note that this response should auto-fill with the correct ID.

6.1 Part 1

Section 1: Pre-experiment instruction

Overview. Thank you for taking part in this online study.

Part 1 (today) and Part 2 (tomorrow). This study consists of two parts. Part 1 takes place now, and Part 2 will take place tomorrow at around the same time. Part 1 and Part 2 are paid separately, and you will receive a participation payment for each part. By completing both parts of the study, you will also have a chance to earn an additional bonus of \$30. Please complete each part in one sitting, without taking breaks in between, and keep your attention until the end.

Comprehension checks. On next pages, you will receive detailed instruction and information about the experiment. Afterwards, you will be asked to complete a few comprehension check questions. You can only participate in the experiment if you diligently complete all comprehension check questions!

Part 1 – Instruction. In this experiment, you will see a hypothetical company with no connection to real-world firms. The company can be either good or bad. A good company means its business goes well, while a bad company means its business goes poorly. The computer decides whether it is good or bad by flipping a fair coin.

- If Heads comes up, the company is good (probability 50%)
- If Tails comes up, the company is bad (probability 50%)

Important: You will not be told the outcome of the coin flip. This means you will not know directly whether the company is good or bad. Instead, you will receive multiple pieces of news about the company.

The news. Imagine that a box contains 100 pieces of news for the company. The computer randomly draws items from this box to show you.

- A good company has a box with 65 positive news items and 35 negative news items.
- A bad company has a box with 35 positive news items and 65 negative news items.

So for a good company, the probability of drawing positive news is 65%; for a bad company it is 35%. After each draw, the news item is placed back into the box, so all draws are **independent and equally informative**.

Summary: You will never know with certainty whether the company is good or bad, but the news is on average helpful. The more positive news you receive, the stronger the indication that the company is actually good.

Communication of the news. For the company, you will receive multiple pieces of news sequentially on separate screens. Each piece of news (positive or negative) will be communicated with an **event** (a short story; an illustrative image may appear alongside but contains no additional information). Events are type-specific: the same event cannot appear

for both positive and negative news, and any event/news combination may appear multiple times. All that matters is how many times you receive positive and negative news.

Your decisions. After the news, you will be asked to decide whether you think the company is good or bad.

Part 1 – Your payment. At the end of Part 1, if your guess is correct, you will earn a \$0.5 bonus (paid after Part 2). If your guess is incorrect, you will not earn the bonus. You will receive a fixed participation payment for completing the study. In addition, if you participate in and complete Part 2 tomorrow, you will receive an additional fixed participation payment and have a chance to win a \$30 bonus.

In the next page, there will be some comprehension checks. You can only continue the experiment if you correctly complete all comprehension check questions.

Section 2: Comprehension checks

[Option sequence is randomized. Seven comprehension check questions; correct answers required to proceed.]

Q1. Is the following statement correct? “The more positive news I receive for a company, the more likely it is that the company is good.” (Yes / No)

Q2. Suppose one event is given by “Factory burns down.” Which statement is correct?

- The event can never occur multiple times.
- The event can occur multiple times as positive and negative news.
- The event can occur multiple times but only as negative news.

Q3. Which of the following statements is correct?

- If I have already seen an event that is associated with positive news, then I know I have already received a positive piece of news for this company.
- The same event can be associated with both positive and negative news.

Q4. Suppose that you have received two negative news items for a company, both communicated with the same event. Which is correct?

- Only one of the two news should be taken into account because they were communicated with the same event.
- I need to take both pieces of news into account because both were randomly and independently determined.

Q5. Suppose that you have received two positive news items for a company. Which is correct?

- I should take them into account more strongly if they were communicated with the same event than if they were communicated with different events.
- I should take them into account less strongly if they were communicated with the same event than if they were communicated with different events.
- It doesn’t matter; I should treat both equally, regardless of whether they come from the same event or different events.

Q6. Suppose that a company is good and that the first news item about this company was positive. What does this imply for the probability that the next piece of news will also be positive?

- The probability is the same as before, since the computer draws independently with replacement.
- The probability is higher than before.
- The probability is lower than before.

Q7. Do images shown alongside the text provide additional information about whether the company is good or bad?

- Yes, the image itself contains important information.
- No, the image is only illustrative; all information comes from the text (story).

Section 3: Group randomization and news delivery

Instruction. You will sequentially receive multiple news about the company on separate screens. You can only review each news once and cannot go back.

Reminder: If your guess is correct at the end of the study you earn a \$0.50 bonus payment.

[Respondents are randomized into one of the following 4 groups.]

- **Group 1:** 18 positive news and 2 negative news.
- **Group 2:** 14 positive news and 6 negative news.
- **Group 3:** 11 positive news and 9 negative news.
- **Group 4 (Stop-and-Think):** 11 positive news and 9 negative news; complete a structured recall task *before* belief elicitation.

News sequences across treatment groups

The table below presents the sequence of 20 news signals assigned to each group. Group G4 receives the same news sequence as G3.

Position	G1	G2	G3
1	P	P	N
2	P	P	P
3	P	N	N
4	P	P	P
5	N	P	P
6	P	P	N
7	P	P	P
8	P	N	N
9	P	P	P
10	P	P	P
11	P	P	N
12	N	N	P
13	P	P	N
14	P	P	P
15	P	N	N
16	P	P	P
17	P	N	N
18	P	P	P
19	P	P	N
20	P	N	P

Example of a positive news item:



The company successfully launched a new product line that quickly attracted strong customer demand. The rollout went smoothly, and sales exceeded expectations in the first months.

The news for the company is positive.

Example of a negative news item:



A major customer cancelled its contract with the company, citing repeated delivery delays and quality concerns. The cancellation will significantly reduce the company's sales and production in the coming months.

The news for the company is negative.

Q8. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? (Yes / No)

Section 4: First belief elicitation and recall

Branch A: If assigned to Groups 1–3.

Q9. Your decision. *[Option sequence is randomized.]* Based on the news items you just received about the company, what is your best guess about the type of this company? (Good / Bad) If your guess is correct, you will earn a \$0.50 bonus.

[Randomly shown one of the questions below.]

Q10v1. What do you think is the probability that the company is “good”? Slider 0–100.

Q10v2. What do you think is the probability that the company is “bad”? Slider 0–100.

Q11. Recalls. *[Sequence is randomized.]* How many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Branch B: If assigned to Group 4.

Q9. Recalls. *[Sequence is randomized.]* How many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Q10. Your decision. *[Option sequence is randomized.]* Based on the news items you just received about the company, what is your best guess about the type of this company? (Good / Bad) If your guess is correct, you will earn a \$0.50 bonus.

[Randomly shown one of the questions below.]

Q11v1. What do you think is the probability that the company is “good”? Slider 0–100.

Q11v2. What do you think is the probability that the company is “bad”? Slider 0–100.

Background information

Q12. What is the state of your primary residence? *[Dropdown list]*

Q13. How would you describe your present employment situation?

- Working full-time (for someone or self-employed)
- Working part-time (for someone or self-employed)
- Not working, but would like to work
- Not actively looking for work
- Student, at school or in training
- Other (please specify) _____

Q14. Which of the following best characterizes your household?

- Own our house/apartment without a mortgage
- Own our house/apartment and have a fixed-rate mortgage
- Own our house/apartment and have a variable-rate mortgage
- Rent our house/apartment

- Other

Q15. Which category represents your total household pre-tax income during the past 12 months?

- Less than \$10,000
- \$10,000 to \$19,999
- \$20,000 to \$29,999
- \$30,000 to \$39,999
- \$40,000 to \$49,999
- \$50,000 to \$59,999
- \$60,000 to \$74,999
- \$75,000 to \$99,999
- \$100,000 to \$149,999
- \$150,000 to \$199,999
- \$200,000 or more

Q16. Does your household hold any stock market investments? (Yes / No)

Q17. What is your gender? (Male / Female / Other / Prefer not to answer)

Q18. What is your age? *[Dropdown list]*

Q19. What is the highest level of school you have completed?

- Less than high school
- High school diploma
- Some college but no degree
- Bachelor's Degree or higher

6.2 Part 2 (one day later)

Welcome

Thank you for participating in this study. This study consists of two parts. Part 1 took place yesterday and Part 2 takes place now. To complete the study, you will need to read all instructions carefully.

Part 2 – Instruction. Yesterday in Part 1, you received several pieces of news about a hypothetical company. This means all of the information you received in Part 1 yesterday is relevant for making your decisions on the following screens. On the next screen, we will ask you to recall the different types of news you received about the company and then make a guess about the company's type.

Part 2 – Payment. At the end of Part 2, if your guess for the company is correct, you will earn a \$0.5 bonus. If your guess is incorrect, you will not earn the bonus. You will receive a fixed participation payment for completing this part. In addition, you have a chance to win an additional bonus of up to \$30.

Section 2: Second belief elicitation and recall

Branch A: If assigned to Groups 1–3.

Q1. Your decision. *[Option sequence is randomized.]* Based on the news items you received yesterday, what is your best guess about the type of this company? (Good / Bad) If your guess is correct, you will earn a \$0.50 bonus.

[Randomly shown one of the questions below.]

Q2v1. What do you think is the probability that the company is “good”? Slider 0–100.

Q2v2. What do you think is the probability that the company is “bad”? Slider 0–100.

Q3. Recalls. *[Sequence is randomized.]* How many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Branch B: If assigned to Group 4.

Q1. Recalls. *[Sequence is randomized.]* How many times was positive news shown to you, and how many times was negative news shown to you?

- Positive news: _____
- Negative news: _____

Q2. Your decision. *[Option sequence is randomized.]* Based on the news items you received yesterday, what is your best guess about the type of this company? (Good / Bad) If your guess is correct, you will earn a \$0.50 bonus.

[Randomly shown one of the questions below.]

Q3v1. What do you think is the probability that the company is “good”? Slider 0–100.

Q3v2. What do you think is the probability that the company is “bad”? Slider 0–100.

7 Quantitative Belief Study (Experiments 4–5)

7.1 Part 1

Section 1: Instruction Overview

Thank you for taking part in this online study.

Part 1 (today) and Part 2 (tomorrow). This study consists of two parts. Part 1 takes place now, and Part 2 will take place tomorrow at around the same time. Part 1 and Part 2 are paid separately, and you will receive a participation payment for each part. By completing both parts of the study, you will also have a chance to earn an additional bonus of \$30.

Instruction. In the next section, you will see a series of **data points generated from a stable process**, presented one at a time. Please review each screen carefully. You do not need to take notes—simply view the information as it is presented. Some of your responses in this section will be selected for a bonus payment. Please read the information carefully, as accurate answers may increase your bonus.

Section 2: Data display

[Respondents are randomized into one of 3 groups. The 20 data points are displayed sequentially in separate screens.]

Group 1: 20 data points from an AR(1) process with $\mu = 5$, $\rho = 0.1$, $\sigma = 0.5$:

5.69, 5.62, 4.97, 5.40, 5.38, 5.47, 4.90, 5.02, 4.65, 4.77, 4.60, 5.21, 5.55, 5.29, 4.75,
5.97, 5.20, 5.28, 5.03, 4.17

Group 2: 20 data points from an AR(1) process with $\mu = 5$, $\rho = 0.9$, $\sigma = 0.5$:

4.19, 4.13, 4.14, 4.17, 5.26, 5.37, 5.05, 5.60, 5.90, 6.47, 6.12, 4.94, 5.45, 5.11, 5.74,
5.20, 5.34, 5.17, 5.02, 4.56

Group 3: 20 data points from an AR(1) process with $\mu = 5$, $\rho = 0.9$, $\sigma = 5$:

17.45, 11.54, 7.38, 11.37, 8.14, 2.37, -2.26, 8.06, 10.33, 4.65, 1.18, -3.32, 0.35,
5.88, 3.00, 2.36, 2.35, 4.96, 6.63, 0.51

Q1. Notes check

Did you write down or take pictures/screenshots/recordings of any of the information provided to you on the previous screens? Your answer will not affect your bonus payment for this study. Please answer truthfully.

- Yes
- No

Q2. Attention check

This is a question to check whether you are paying attention and reading the questions carefully. Please select both “strongly disagree” and “strongly agree” to continue. Otherwise, you will leave the survey.

- Strongly agree
- Agree
- Neither strongly agree or disagree
- Disagree
- Strongly disagree

Section 3: Background information

Q3. What is the state of your primary residence? *[Dropdown list]*

Q4. How would you describe your present employment situation?

- Working full-time (for someone or self-employed)
- Working part-time (for someone or self-employed)
- Not working, but would like to work
- Not actively looking for work
- Student, at school or in training
- Other

Q5. Which category represents your total household pre-tax income during the past 12 months?

- Less than \$10,000
- \$10,000 to \$19,999
- \$20,000 to \$29,999
- \$30,000 to \$39,999
- \$40,000 to \$49,999
- \$50,000 to \$59,999
- \$60,000 to \$74,999
- \$75,000 to \$99,999
- \$100,000 to \$149,999
- \$150,000 to \$199,999
- \$200,000 or more

Q6. Does your household hold any stock market investments? (Yes / No)

Q7. What is your gender? (Male / Female / Other / Prefer not to answer)

Q8. What is your age? *[Dropdown list]*

Q9. What is the highest level of school you have completed, or the highest degree you have received?

- Less than high school
- High school diploma
- Some college but no degree
- Bachelor's Degree or higher

Section 4: Estimate and recalls

Q10. Your estimate. Based on the data you just observed, what is your best estimate of the **average of the series of data**?

[Bonus: After the study, answers will be compared to the true average of the data series you just observed. The 10% of participants with the most accurate answers will each receive a \$5 bonus.]

Average of the series: _____

Q11. Recalls. Please enter any values from the data series that you remember seeing earlier. You may enter as many values as you can remember.

- Data observation #1: _____
- Data observation #2: _____
- Data observation #3: _____
- Data observation #4: _____
- Data observation #5: _____
- Data observation #6: _____
- Data observation #7: _____
- Data observation #8: _____
- Data observation #9: _____
- Data observation #10: _____

Q12. Do you need more space? (Yes / No)

Q13. *[Display if Q12 = Yes]*

- Data observation #11: _____
- Data observation #12: _____
- Data observation #13: _____
- Data observation #14: _____
- Data observation #15: _____

- Data observation #16: _____
- Data observation #17: _____
- Data observation #18: _____
- Data observation #19: _____
- Data observation #20: _____

7.2 Part 2 (one day after Part 1)

Welcome

Thank you for participating in this study. This study consists of two parts. Part 1 took place yesterday and Part 2 takes place now. To complete the study, you will need to read all instructions carefully.

Part 2 – Instruction. Yesterday in Part 1, you received a series of data points generated from a stable process. This means all of the information you received in Part 1 yesterday is relevant for making your decisions on the following screens. You have a chance to win an additional bonus of up to \$30 by completing this part of the survey.

Q1. Your estimate

Based on the data you observed yesterday in Part 1, what is your best estimate of the average of the series of data?

[Bonus: After the study, answers will be compared to the true average of the data series you received yesterday in Part 1. The 10% of participants with the most accurate answers will each receive a \$5 bonus.]

Average of the series: _____

Q2. Recalls

Please enter any values from the data series that you remember seeing yesterday in Part 1. You may enter as many values as you can remember.

- Data observation #1: _____
- Data observation #2: _____
- Data observation #3: _____
- Data observation #4: _____
- Data observation #5: _____
- Data observation #6: _____
- Data observation #7: _____
- Data observation #8: _____
- Data observation #9: _____

- Data observation #10: _____

Q3. Do you need more space? (Yes / No)

Q4. *[Display if Q3 = Yes]*

- Data observation #11: _____

- Data observation #12: _____

- Data observation #13: _____

- Data observation #14: _____

- Data observation #15: _____

- Data observation #16: _____

- Data observation #17: _____

- Data observation #18: _____

- Data observation #19: _____

- Data observation #20: _____

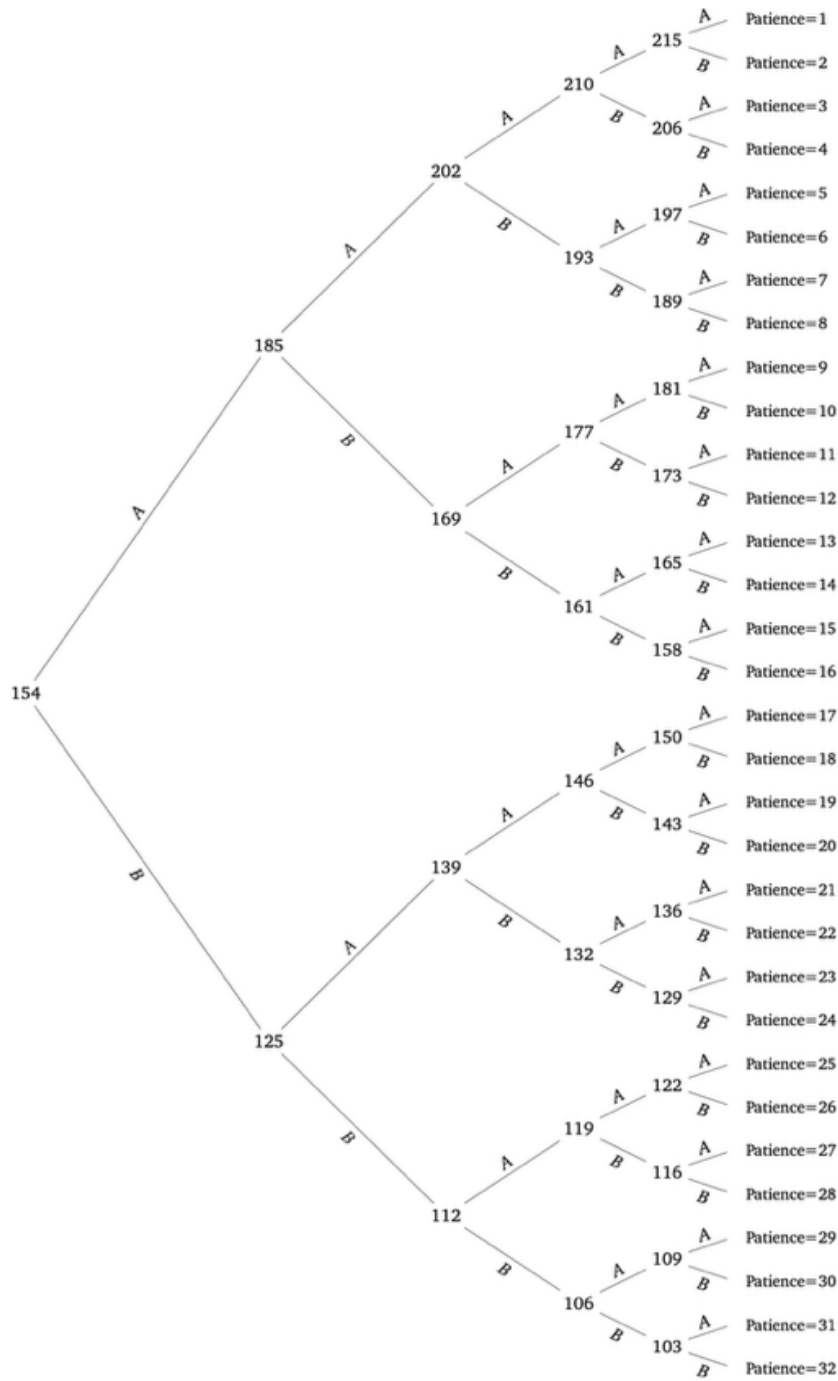


Figure 1: Staircase time-preference task. A = choice of “100 dollars today,” B = choice of “x dollars in 12 months.” Each respondent is first asked whether they prefer 100 dollars today or 154 dollars in 12 months. If the respondent chooses A, the future payment increases on the next question; if B, it decreases. The procedure iterates over five questions.

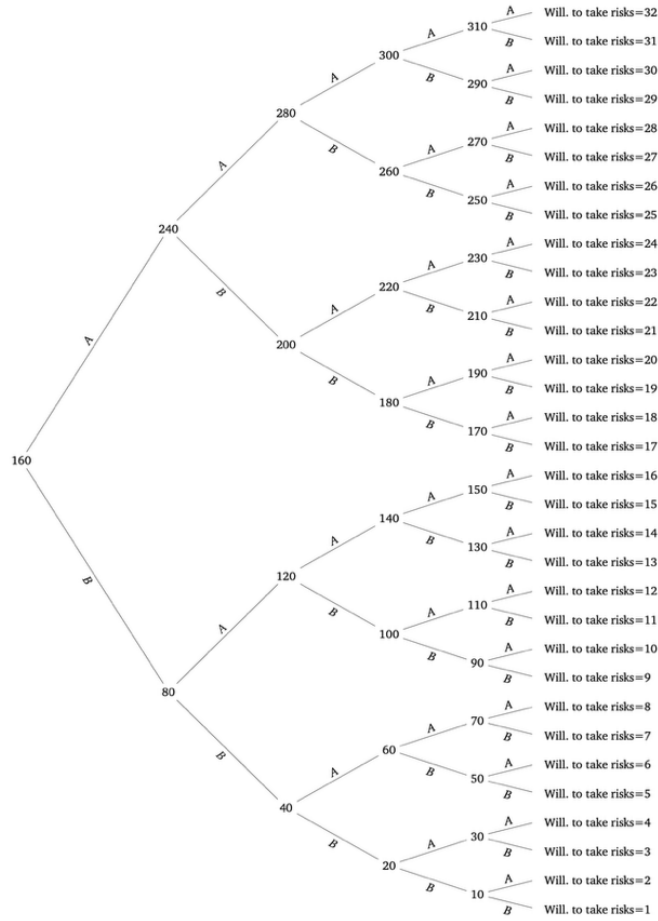


Figure 2: Staircase risk task. A = choice of the lottery (50:50 chance of \$300 or nothing); B = choice of the sure payment. Each respondent is first asked whether they prefer 160 dollars for sure or the lottery. If B, the safe amount in the next question decreases; if A, it increases. The procedure iterates over five questions.